

Trainings ▾

General Information

Cooling system **must** be cleaned when one or more of the following contaminants are found in the cooling system:

- Silicate gel
- Oil, grease, or fuel
- Scale
- Rust
- Solder bloom.

These conditions may be accompanied by:

- Coolant discoloration
- Presence of sediment in coolant samples drawn from radiator or a low point in cooling system
- Engine overheating.

To remove lubricating oil or fuel contamination from a cooling system, a low foaming cleaner specifically designed for oil removal **must** be used. Fleetguard Restore™ Heavy-Duty Cleaner is an alkaline based product modified to perform as a lubricating oil and grease super cleaner, and can also remove silicate gelation from a cooling system.

For cleaning poorly maintained or severely contaminated cooling systems, an acid based cleaner is recommended. Fleetguard Restore™ Plus Heavy-Duty Cleaner is an acid based product that is excellent for removing rust, scale, solder bloom, and other corrosion contaminants from the cooling system.

Alternative cleaning products can be used, however the use of products containing hydrochloric acid will **not** result in adequate system cleanup and may attack cooling system materials. Additionally, the use of liquid dish soap is **not** authorized. Foaming and air lock in the cylinder head(s) can occur causing severe engine damage. Prolonged use of any cleaner longer than 3 hours is **not** recommended.

Table 1 below lists various contaminants and the cleaning performance of each cleaner with respect to those contaminants. The correct cleaner **must** be chosen based on the type of contamination.

Table 1 - Cooling System Cleaner Application Chart

Contaminant	Fleetguard® Restore™ (Alkaline), Part Number CC2610, CC2611, or CC2612, or Equivalent Product	Fleetguard® Restore Plus™ (Acid), Part Number CC2637, Part Number CC2638, or Equivalent Product
Silicate gel	Excellent	Poor
Oil, grease, and fuel	Excellent	Good
Scale	Poor	Excellent
Rust	Poor	Good
Solder bloom	Poor	Good

Cooling systems **must** be cleaned carefully when any of the above conditions in Table 1 are apparent. Flushing the cooling system of chemical cleaners completely and thoroughly at the conclusion of the cleaning process is critical. Flushing **must** be done with water, and will require more than one flushing to purge the cooling system of the cleaning chemicals. The cooling system **must** be flushed with minimum volume of water, three times the total volume of the cooling system.

Failure to purge cleaning chemicals sufficiently can result in contamination of the new coolant, which can lead to additive depletion, severe corrosion, and engine damage. Reference Procedure 008-018 in Section 8 of the corresponding Service Manual for cooling system cleaning procedure.